5V Adapter Input

J3
10104110-0001LF
VBUS
Shield
GND
D+
D-
ID
EXT_5V
GND

USB Data / Flashing

J1
10104110-0001LF
VBUS
Shield
GND
D+
D-
ID
D+
D-
GND

Control & Processing Unit

3V3
R1
10 K-Ohms
SW1
RST
C1
1uF
GND
R2
SW2
BOOT
DFR0026IN
SEN0244
C3
100nF
pH-4502CIN
RAIL_SENSE
R8
4.7 K-Ohms
DS18B20
RelayIN
U1
ESP32-S3-WROOM-1
EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
PSRAM
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
TRIG_HC-SR04
R14
15 K-Ohms
10 K-Ohms
R15
ECHO_HC-SR04
R3
22
D-
R4
22
D+
I2C_SDA
I2C_SDA
R7
4.7 k-Ohms
I2C_SCL
I2C_SCL
R6
4.7 k-Ohms

DS18B20 Connector

J4
DS18B20
DS18B20
3V3
GND

GPIO8

10 K-Ohms
EXT_5V
R13
RAIL_SENSE
R11
6.8 K-Ohms
R12
6.8 K-Ohms
GND

BME280 Connector

J2
BME280
3V3
GND
I2C_SCL
I2C_SDA

HC-SR04 Connector

J7
HC-SR04
GND
ECHO_HC-SR04
TRIG_HC-SR04
EXT_5V

pH-4502C

J14
pH-4502C
GND
GND
EXT_5V
R10
10k
R9
15k
C4
1uF
pH-4502CIN
GND

SEN0244 Connector

J5
SEN0244
SEN0244
EXT_5V
GND

SGP30 Connector

J11
SGP30
3V3
GND
I2C_SCL
I2C_SDA

Mounting Holes

H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole

Battery Mangement System

J10
MRB045
12V
GND
J6
Battery Power

Power Switching Circuit

12V
F1
Fuse
R16
1 K-Ohms
GND
J9
MP1584 5V_Battery
GND
R19
220 Ohms
EXT_5V
Q2
IRLZ44N
R18
10 K-Ohms
GND
R17
100 K-Ohms
GND
Q1
IRLZ44N
D1
1N5822
D2
1N5822
EXT_5V
GND
3V3
GND
J12
MP1584 3V3
GND

3V3 Power Branch

3V3
C7
10nF
C8
470uF
C9
10uF
C10
100nF
3V3
C11
100nF
3V3
C12
100nF
3V3
C13
100nF
3V3
C14
100nF
3V3
C15
1uF
GND

5V Power Branch

EXT_5V
C16
470uF
C17
10uF
C18
100nF
GND
EXT_5V
C22
100nF
C23
1uF
C24
10uF
GND
EXT_5V
C19
470uF
C20
10uF
C21
100nF
GND
EXT_5V

Relay Control Connector

J13
Relay
GND
RelayIN
EXT_5V

DFR0026

DFR0026
J8
3V3
GND
C2
10uF
GND
R5
10 K-Ohms
GND
DFR0026IN

I-ACT
Sheet: /
File: PCB Design.kicad_sch
Title: PCB Design
Size: A3
Date: 2026-11-05
Rev: 1.5
KiCad E.D.A. 10.0.8
Id: 1/1